

SHREESH JUGADE

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EDUCATION

Pune Institute of Computer Technology

Pune, India

Bachelor of Technology in Computer Engineering (GPA: 9.52)

August 2024 – May 2028

- Minor in Financial Technology & Management
- Coursework: Data Structures & Algorithms, Data Warehousing and Mining, Artificial Intelligence, Machine Learning, Database Management Systems, Operating Systems, Computer Networks

PROJECTS

Pre-Delinquency Risk Prediction Platform | Python, PyTorch, XGBoost, LightGBM, CatBoost, FastAPI, Redis

[GitHub](#)

- Built a credit-risk prediction platform combining XGBoost, LightGBM, CatBoost and PyTorch LSTM models to capture temporal financial behavior and nonlinear risk patterns, with SHAP-based explainability.
- Engineered temporal behavioral features including lag, delta, rolling-statistical, volatility, trend, and persistence features, and integrated them into a FastAPI inference service exposing REST APIs for real-time and batch financial risk scoring.
- Optimized concurrent CPU-bound inference using ProcessPoolExecutor multiprocessing and integrated Redis caching with token-bucket rate limiting, achieving <50 ms cached latency and 1,400+ requests/sec under Locust load testing.

In-Memory Vector Database | C++, HNSW, ANN Search

[GitHub](#)

- Developed an in-memory vector database in C++ for high-dimensional embedding storage and similarity search, implementing brute-force k-NN, greedy graph search, and Hierarchical Navigable Small World (HNSW) indexing.
- Applied an optimized HNSW index using multi-layer graph traversal, priority-queue candidate exploration, and tunable efConstruction/efSearch parameters to improve retrieval accuracy and query efficiency at scale.
- Benchmarked retrieval on 100K 128-dimensional vectors using Euclidean distance, achieving 1.29 ms average query latency, an 81% reduction compared with exhaustive linear search.

Type 2 Diabetes Clinical Trial Pre-Screening Agent | Python, LangGraph, Gemini 2.0 Flash, RAG, Regex

[GitHub](#)

- Built a 4-stage LangGraph agent for Type 2 Diabetes clinical-trial pre-screening, orchestrating structured filtering, evidence retrieval, deterministic criterion evaluation, and coordinator-facing report generation across patient and trial data.
- Implemented RAG-based evidence retrieval with structured patient data and regex-based extraction of HbA1c/eGFR thresholds, evaluating age, laboratory values, medications, and recruiting status with explicit uncertainty and conflict states.
- Developed a 12-case automated evaluation suite spanning retrieval, criterion correctness, agent behavior and output traceability, achieving 85% eligibility-threshold parsing coverage across the clinical-trial dataset.

PUBLICATIONS

Jugade, S., Santre, M., Shevde, A., Panhalkar, A. and Bapat, N., “MAITRI: A Multimodal AI System for Psychological and Physical Well-Being of Astronauts.” Presented at the *2026 IEEE Space, Aerospace and Defence Conference (SPACE)*.

TECHNICAL SKILLS

Languages: C/C++, Python, Java, Go, JavaScript, SQL, Shell Scripting

AI/ML: LLMs, PyTorch, LangGraph, RAG, XGBoost, LightGBM, CatBoost

Frameworks & Libraries: FastAPI, WebSockets, Redis, PostgreSQL

Developer Tools: Git, Docker, Linux, MongoDB, Postman

LEADERSHIP & EXTRACURRICULARS

Operations Head, PICT's annual flagship event INC'26: Co-led a team of 40 volunteers and coordinated logistics, scheduling, and communication for 150+ judges and guest speakers during a two-day international techfest.

Winner, PICT TechRush Hackathon 2025: Developed a weighted ensemble using XGBoost and LightGBM to predict solar power generation from weather and atmospheric features as part of the winning team.

Volunteer Instructor, PICT Model School: Conducted an interactive AI awareness session for school students, covering core AI concepts, real-world applications, and responsible use of AI, with examples demonstrating how AI is used in everyday life.